# PHY · Wave–particle duality and quantum theory — syllabus points and past-paper evidence

Every award of the 2016–2025 papers, labelled to the 2026/2027 Year 12 syllabus content (identical in both). Statistics from the labels; notes written by AI from that evidence: review before use.

## Pattern 1: Diffraction and interference (Young's experiment)

Syllabus: S26-U4-SU-23 / S27-U4-SU-23

Frequency: 5 of 10 exam years; 29 marks in all.

Marks reward: Qualitative explanation 14, Final answer 6, Select the principle or relationship 3, Recall 2, Graph work 2, Evaluate and conclude 1, Substitution and working 1

Past questions: 2016P-20(e); 2016R-11(a); 2018R-6(a); 2018R-6(b); 2019P-17(b); 2019P-17(c); 2020R-8(a); 2020R-8(b); 2025P-13(b); 2025P-13(ci); 2025P-13(cii); 2025P-13(d)

Summary: This point asks for the wave evidence for light, mainly diffraction and interference, most often through Young's double-slit set-up. Students explain how the fringe pattern forms and sometimes calculate fringe spacing or wavelength from it. Past papers lean heavily on written explanation, with smaller calculation and graph-reading parts.

Recognise: A question mentions a double slit, a diffraction pattern, bright and dark fringes, an interferometer or a diffraction grating, or asks what property of light a pattern shows. Calculation versions give a fringe spacing, slit separation and screen distance, or ask for a gradient from a graph of fringe spacing against another variable.

### Steps

1. Name the phenomenon: diffraction is the spreading of waves at slit edges, and interference is the overlap of those spread waves on the screen.
2. Link brightness to phase: waves arriving in phase reinforce to give a bright fringe, while waves 180 degrees out of phase cancel to give a dark fringe.
3. Relate this to path difference: a bright fringe needs path difference = mλ and a dark fringe needs path difference = (m + 1/2)λ, where m = 0, 1, 2 and so on.
4. Use fringe spacing s = λD / d, where d is slit separation and D is the slit-to-screen distance. Rearrange it for the unknown and keep units consistent.
5. For a graph question, take the gradient from the line of best fit, then use the gradient relationship (gradient = λD/d or λ/d as set up in the question) to find the wavelength.
6. For a change question, identify which quantity changes (for example the wavelength of the particles or light), then state the direction using the proportionality in s = λD/d.
7. For electrons or other particles, find the wavelength from the de Broglie relation first, then explain that one wavelength is needed for clear fringes.

### Where the marks are

- Name diffraction or interference as the wave evidence; saying only that light is a wave is not enough.
- Give a causal chain for the pattern: spreading at slits, overlap on the screen, in phase or out of phase, then bright or dark fringe.
- Use the fringe spacing relationship correctly, showing the rearrangement and consistent units.
- For graph questions, draw construction lines on the line of best fit, read the gradient, then connect it to wavelength.
- For evaluation questions, state a clear conclusion that the pattern supports a wave model of light.

### Pitfalls

- Describing light as bending or reflecting at the slits instead of diffracting, or stating the pattern without giving the reason for it.

- Mixing up slit separation and screen distance in s = λD/d, or missing a unit conversion from nm or mm to m.
- Giving the wrong phase or path condition for dark fringes, such as using a whole-number multiple of the wavelength.
- Reading the gradient from two data points instead of two points on the line of best fit.
- Answering a change question with a direction alone, without the proportionality that justifies it.

## Pattern 2: Light is not a mechanical wave

Syllabus: S26-U4-SU-24 / S27-U4-SU-24

Frequency: 2 of 10 exam years; 4 marks in all.

Marks reward: Recall 3, Qualitative explanation 1

Past questions: 2017R-1 ; 2021P-16(d)

Summary: This point asks you to show that light behaves like a wave in many ways but is not a mechanical wave, because it can travel through a vacuum. Questions are short and usually ask for a conclusion from a scenario, such as light arriving from the Sun, or for named wave phenomena that support the wave model.

Recognise: Look for wording such as 'light travelling from the Sun to Earth', 'through a vacuum', 'does not need a medium', or 'what conclusion can be drawn'. A second form asks you to list phenomena that support the wave nature of light, usually refraction, dispersion, diffraction, interference or polarisation.

Steps

1. Recall the defining feature of a mechanical wave: it transfers energy by making particles of a medium vibrate, so it needs a material medium.
2. Look at the scenario and ask whether there is a medium between the source and the observer. Sunlight crosses the near-empty space between the Sun and Earth, which is a vacuum.
3. Link the two ideas: because light can travel with no medium present, it cannot be modelled as a mechanical wave.
4. State the conclusion in one clear sentence, for example that light is not a mechanical wave (it can be treated as an electromagnetic wave or as a particle-like carrier of energy).
5. For a question asking for supporting phenomena, name wave behaviours that light shows, such as refraction (change in speed and direction at a boundary) and dispersion (splitting into colours because the refractive index varies with wavelength). Give each one separately.

Where the marks are

- One mark for stating the key reason: light can travel through a vacuum, so there is no medium.
- One mark for the conclusion itself: light is not a mechanical wave (or it behaves as a particle when this is the explanation being tested).
- One mark for each correctly named wave phenomenon when a list is requested; a single correct name earns only its own mark, so give two distinct phenomena.
- Credit is given for a conclusion stated in words, so a bare label without a reason may lose the explanation mark.

Pitfalls

- Saying that light needs air or an ether to travel, which reverses the key idea.
- Giving a conclusion such as 'light is different' without saying it is not a mechanical wave.
- Stating the vacuum fact but never drawing the conclusion from it.
- Listing particle-only evidence, such as the photoelectric effect, as support for light being a wave.
- Repeating a phenomenon in two words that mean the same thing, which earns only one mark.

## Pattern 3: Polarisation and the transverse wave model

Syllabus: S26-U4-SU-25 / S27-U4-SU-25

Frequency: 2 of 10 exam years; 18 marks in all.

Marks reward: Qualitative explanation 6, Recall 4, Final answer 4, Substitution and working 4

Past questions: 2016P-20(e); 2023C-20(a); 2023C-20(b); 2023C-20(ci); 2023C-20(cii); 2023C-20(d); 2023C-20(ei); 2023C-20(eii)

Summary: This point asks why polarisation shows that light is a transverse wave: only a transverse wave has oscillations that can be restricted to one plane. Questions usually ask students to name polarisation (or another wave-only effect) and explain it, or to apply Malus' law to a filter or a stack of filters.

Recognise: Look for polarising filters or sheets, a 'polarisation axis' or 'axis of the filter', intensity changing as a filter is rotated, crossed filters with light getting through a third filter placed between them, or wording like 'demonstrates wave behaviour' or 'use Malus' law'. Any question that asks what a filter does to an unpolarised beam is on this point.

## Steps

1. Identify what is being asked: a description of the mechanism, a named wave effect, a calculation with Malus' law, or the intensity after several filters.
2. For a mechanism, say the filter (for example long polymer chains, or a material that absorbs one orientation of the electric field) lets through only the component of the electric field parallel to its axis, and absorbs or blocks the component perpendicular to it.
3. Link the result to the transverse model: the electric field oscillates perpendicular to the direction of travel, so its orientation can be fixed in one plane. A longitudinal wave has no such orientation, so it could not be polarised.
4. For Malus' law, write I = I0 cos^2(theta), where theta is the angle between the light's polarisation direction and the filter axis. Use the given fraction of intensity to find cos^2(theta), then take the square root and the inverse cosine.
5. For a stack of filters, multiply the transmitted fraction at each stage. The first filter takes unpolarised light down to half intensity, then each following filter applies its own Malus' law factor with the angle between successive axes.
6. Check the answer against the physics: intensity should not exceed the value after the previous filter, and the angle should give the stated fraction of intensity.

## Where the marks are

- A mark for naming a wave-only effect such as polarisation, interference or diffraction, and for saying it shows the wave is transverse.
- Marks for explaining the filter mechanism in order: the electric field component parallel to the axis is absorbed or transmitted, and only the perpendicular component passes through.
- Marks for defining the filter axis correctly, meaning the direction that lets the electric field component through, and for describing its effect on passing light.
- Separate marks for each step of a Malus' law calculation: substituting the intensity ratio, cancelling I0, taking the square root, and giving the angle with the correct units.
- For stacked filters, marks for using the angle between successive axes, multiplying the fractions and converting to a percentage when asked.

## Pitfalls

- Writing that the filter 'blocks' the wave without saying which component of the electric field is absorbed or transmitted.
- Confusing the filter axis with the direction of the chains or with the direction of the wave's travel.
- Using the intensity ratio where the amplitude ratio is needed, or forgetting to square the cosine in Malus' law.
- Taking the angle as the complement of the answer, or dropping the inverse cosine step, so the angle returned is the cosine value.
- Multiplying the filter fractions incorrectly when a third filter is added, or failing to convert a fraction to a percentage when the question asks for one.

## Pattern 4: Electromagnetic waves

Syllabus: S26-U4-SU-26 / S27-U4-SU-26

Frequency: 3 of 10 exam years; 4 marks in all.

Marks reward: Recall 2, Final answer 1, Qualitative explanation 1

Past questions: 2017R-10； 2021C-20(eiii)； 2023C-20(a)

Summary: This point covers the structure of electromagnetic (EM) waves: they are transverse, with an electric field and a magnetic field that oscillate at right angles to each other and to the direction of travel. It appears in the papers as short knowledge or description items, often embedded in a context such as antennas, light or the EM spectrum, and sometimes as a follow-on identification question.

Recognise: Look for wording about how EM waves are produced or described (for example, by an oscillating charge), questions asking what a wave consists of or how its fields are oriented, and questions that link a wavelength or frequency to a region of the EM spectrum. Questions about polarisation of light or how a receiver detects a signal often rely on the same transverse-field idea.

Steps

1. State the key fact: an EM wave is transverse, made of an electric field E and a magnetic field B that oscillate in phase with each other.
2. Show that the two fields are mutually perpendicular to each other and both perpendicular to the direction of propagation.
3. If the question asks how the wave is produced, link it to an accelerating or oscillating charge, which sets up the changing electric and magnetic fields.
4. If the question asks how a receiver picks it up, explain that the wave's oscillating electric field exerts a force on charges in the receiving conductor, setting them in motion.
5. For a spectrum-identification question, convert the given value using $c = f\lambda$, then compare the wavelength or frequency with the approximate ranges of the spectrum regions and name the matching region.

Where the marks are

- Full marks for the transverse definition require both fields named, both stated as oscillating, and both stated as mutually perpendicular.
- Explanations of detection or polarisation get a mark for linking the electric field to a force on charges or electrons.
- Spectrum identification earns a mark for naming a region that is consistent with the candidate's own earlier calculated value, even when the region is not the one a rounded value would suggest.
- Describing production as an oscillating charge is worth a mark when the question asks how the wave is made.

Pitfalls

- Saying the fields are parallel or in the same plane, or failing to say they are perpendicular to each other.
- Describing the wave as longitudinal, or stating only that it is a wave without saying what oscillates.
- Naming only one field, or describing a field without saying it oscillates.
- In the spectrum question, quoting a region without checking it against the wavelength or frequency value previously calculated in the same question.
- Writing vague detection answers such as the receiver just absorbs the wave, without saying that the electric field acts on charges.

## Pattern 5: Oscillating charges and the electromagnetic spectrum

Syllabus: S26-U4-SU-27 / S27-U4-SU-27

Frequency: 2 of 10 exam years; 3 marks in all.

Marks reward: Qualitative explanation 3

Past questions: 2017R-10;  2018R-8(a)

Summary: This point covers the two-way link between an oscillating charge and electromagnetic waves: a charge oscillating at some frequency radiates an EM wave of that same frequency, and an incoming EM wave makes charges oscillate at its frequency. It usually appears as a short explain or describe question, often in an antenna or transmission-and-reception context, where students must describe the wave produced and how it is detected.

Recognise: Look for questions about how a radio or microwave antenna transmits and receives signals, how an electric charge produces light or radio waves, or why a receiver can detect a broadcast. Key cue words include oscillating or vibrating charge, accelerating charge, electromagnetic wave, frequency, and receiver or antenna picking up a signal. Questions on the electromagnetic spectrum may also rely on the idea that the wave's frequency is set by the source.

Steps

1. Identify the source: the charge is oscillating (accelerating back and forth), so it is a source of electromagnetic radiation.
2. State the wave produced: an electromagnetic wave whose frequency equals the frequency of the charge's oscillation. The source sets the frequency, and it does not change as the wave travels into a new medium.
3. If the question involves wavelength, use c = f x wavelength; in air or vacuum c is about 3.0 x 10^8 m/s, so a higher oscillation frequency gives a shorter wavelength.
4. Describe reception: the oscillating electric field of the incoming wave exerts a force on free electrons in the receiving antenna, making them oscillate at the same frequency as the wave.
5. Link the two directions explicitly: the oscillating charge sends a wave at its own frequency, and the wave drives charges at the receiver at that frequency. A tuned circuit selects the wanted frequency from the induced oscillation.

Where the marks are

- A mark for stating that the wave produced has the same frequency as the charge's oscillation.
- A mark for stating that the wave makes charges in the receiver oscillate at the wave's frequency; full marks usually need both directions of the link.
- A mark for a cause-and-effect sentence: the oscillating charge emits the wave, and the wave is not the motion of the charge itself.
- If the question covers both transmitting and receiving, each end should be described separately rather than only one end.

Pitfalls

- Believing the wave's frequency differs from the oscillation frequency, or saying the frequency is set by the medium.
- Describing only the transmitting end and forgetting to explain how the receiving charges are made to oscillate.
- Confusing electromagnetic waves with mechanical waves such as sound, for example saying a medium or particle transfer is needed.
- Writing only that the receiver 'picks up the signal' without stating the physical mechanism of electrons oscillating at the same frequency.
- Mixing up charge oscillation with current, or saying the charge itself travels along the antenna to the receiver.

## Pattern 6: Quantised states of matter and energy

Syllabus: S26-U4-SU-28 / S27-U4-SU-28

Frequency: 1 of 10 exam years; 2 marks in all.

Marks reward: Qualitative explanation 2

Past questions: 2017R-12

Summary: This point asks why energy and matter behave as discrete packets rather than continuously. It usually appears as a short explain or describe question built on black body radiation, where classical theory fails at high frequency, with the photoelectric effect and atomic line spectra as other evidence. The papers reward explanation of the evidence more than calculation.

Recognise: Look for black body radiation curves, the ultraviolet catastrophe, a classical prediction that clashes with observation, emission or absorption lines, or a photoelectric threshold. The command words are usually explain, describe or why, and the question asks for reasoning rather than a numerical answer.

Steps

1. State what classical wave theory predicts: energy emitted continuously, with the intensity at short wavelengths (high frequency) rising without limit.
2. Describe what the black body curve actually shows: intensity peaks at a particular wavelength and falls again at high frequency, and the curve depends only on temperature.
3. Name the mismatch between prediction and observation, the ultraviolet catastrophe, and say which region of the graph it occurs in.
4. Conclude that the only way to fit the data is if energy is emitted and absorbed in discrete amounts, called quanta, with each quantum of energy E = hf, where h is Planck's constant and f is frequency.
5. Where the question allows, support the conclusion with other evidence: atomic line spectra show discrete energy levels (photon energy equals the difference between levels), and the photoelectric effect shows that light energy arrives in packets whose energy depends on frequency, not intensity.

Where the marks are

- A mark for stating the classical prediction and that it fails at high frequency, with the ultraviolet catastrophe named or clearly described.
- A mark for concluding that energy is emitted in discrete packets (quanta), which is the reason the curves can be explained.
- A mark for the link between quantum energy and frequency (E = hf) when it is used to support the conclusion.
- Descriptive parts of the same question earn marks for black body features: it absorbs all incident radiation, and its emission depends on temperature alone, not the material.

Pitfalls

- Explaining the graph without stating what classical theory predicted, so the reason quanta are needed is missing.
- Saying the ultraviolet catastrophe means there is too little ultraviolet light; the problem is that classical theory predicts too much at high frequency.
- Writing that energy becomes quantised or light is quantised in general, rather than saying energy is emitted or absorbed in discrete packets.
- Confusing frequency and wavelength on the axes, or reading intensity as frequency when describing the curve.
- Describing a black body as a black-coloured object, instead of a perfect absorber and emitter of radiation.

## Pattern 7: Black-body radiation

Syllabus: S26-U4-SU-29 / S27-U4-SU-29

Frequency: 5 of 10 exam years; 24 marks in all.

Marks reward: Graph work 10, Qualitative explanation 6, Recall 5, Final answer 3

Past questions: 2017R-12; 2017R-5(a); 2017R-5(b); 2017R-5(c); 2018R-8(a); 2018R-8(b); 2022C-18(b); 2024R-2(a); 2024R-2(b); 2025P-16(a); 2025P-16(d)

Summary: This point asks why the observed thermal radiation spectrum of a black body cannot be explained

by the classical wave model, and why light must be treated as quanta (particles) of energy to account for the shape of the spectrum and how it shifts with temperature. Papers usually test it through reading or sketching intensity-against-wavelength graphs at different temperatures, plus a short explanation of the ultraviolet catastrophe and the quantum idea.

Recognise: Look for a black body, incandescent globe or LED described with intensity-against-wavelength curves at several temperatures, or a question asking why the classical model fails. A star's colour linked to its surface temperature, or a request to find a peak wavelength and use the Wien relation, also signals this point.

Steps

1. Define a black body as an ideal object that absorbs all incident radiation without reflecting any, and emits a spectrum that depends only on its temperature.
2. Describe or sketch the graph: as temperature rises, total intensity increases and the peak moves to shorter wavelength (higher frequency). Each curve has one peak and falls towards zero at both ends; label both axes.
3. State the classical prediction: wave theory with energy shared equally among all modes predicts intensity growing without limit at short wavelengths (the ultraviolet catastrophe), which is not observed.
4. Give the quantum resolution: each emitted packet of light has energy E = hf = hc/lambda, so high-frequency modes need a large energy packet and are rarely excited at low temperature. This suppresses the short-wavelength end and produces a peak.
5. Use Wien's displacement law, lambda_max x T = b, to relate peak wavelength and temperature. A straight-line graph of T against 1/lambda_max has gradient b, so read off two points on the line of best fit to calculate it.

Where the marks are

- Marks for a complete black body definition that covers absorbing all incident radiation, no reflection, and re-emission governed by temperature.
- Marks for accurate graph features: the hotter curve has a higher peak at a shorter wavelength, the cooler curve a lower peak at a longer wavelength, and each curve is a smooth single peak that approaches zero at both ends.
- Marks for naming the classical failure precisely: classical theory predicts unlimited intensity at short wavelengths (ultraviolet catastrophe).
- Marks for linking the quantum idea to the observed shape: energy E = hf is emitted in discrete packets, so high-frequency emission is restricted.
- Marks for calculating Wien's constant from the gradient of the line of best fit, showing the two chosen points on the graph and giving the unit (m K).

Pitfalls

- Reversing the trend and saying the peak moves to longer wavelength as temperature rises.
- Drawing curves that do not return to zero at the ends, or drawing peaks that are the same height at every temperature.
- Defining a black body only as 'black' or 'non-reflecting' and omitting that its emission depends on temperature.
- Mixing up gradient and intercept when using T against 1/lambda, or leaving the unit off the calculated constant.
- Stating that classical theory is 'wrong' without identifying the ultraviolet catastrophe or the need for energy quanta, which loses the explanation marks.

## Pattern 8: Photons and E = hf

Syllabus: S26-U4-SU-30 / S27-U4-SU-30

Frequency: 10 of 10 exam years; 104 marks in all.

Marks reward: Qualitative explanation 33, Substitution and working 30, Final answer 16, Select the principle or relationship 13, Recall 5, Graph work 5, Evaluate and conclude 2

Past questions: 2016C-22(b); 2016C-22(d); 2016P-20(a); 2016P-20(d); 2016R-6; 2017P-15(aii); 2017P-18(bi); 2017P-18(ci); 2017P-18(ciii); 2017P-18(d); 2018P-14(d); 2018P-17(d); 2018R-3(b); 2019C-20(d); 2019P-14(b); 2020C-20(bi); 2020C-20(bii); 2020R-3; 2021C-20(a); 2021C-20(eii); 2021C-20(fi); 2021C-20(fii); 2021P-13(d); 2022C-18(a)...

Summary: This point covers the idea that light is emitted and absorbed in discrete packets (photons), with photon energy linked to frequency and wavelength by $E = hf = hc/\lambda$, and to momentum by $E = pc$. It appears often as a short calculation of photon energy, frequency or wavelength (including energy-level transitions, the photoelectric effect and mass-energy problems), usually followed by a written explanation of what the result shows about light as particles.

Recognise: Look for a question giving a wavelength, frequency, energy level change, work function, or mass of a particle and asking for an energy, frequency or wavelength. Questions asking why red light or a change in intensity does or does not eject electrons, or why a photon's wavelength changes after a collision, are also on this point. Words such as photon, threshold, work function, energy level, annihilation, or X-ray fluorescence signal it.

## Steps

1. Write down the given quantities and their units, then convert to SI: nm to m, keV or MeV to J, and eV to J using 1 eV = $1.60 \times 10^{-19}$ J.
2. State the relationship you need: $E = hf$ for a frequency, $E = hc/\lambda$ for a wavelength, and rearrange to $f = E/h$ or $\lambda = hc/E$ as required.
3. For an energy-level transition, find the energy gap $\Delta E$ between the two levels; the largest gap gives the highest frequency and the shortest wavelength.
4. For a photoelectric problem, calculate photon energy $E = hf$, then use $E_{max} = hf - \varphi$, and the threshold frequency $f_0 = \varphi/h$ if asked whether emission occurs.
5. For mass-energy or pair problems, use $E = mc^2$ for the energy needed, then $hf = mc^2$ to find a minimum frequency or wavelength.
6. Give the answer with correct units and sensible significant figures, then check it lies in the expected region of the spectrum (for example UV, X-ray or gamma).
7. For explanation parts, state that energy per photon depends only on frequency, and link the result to a particle view of light (a single photon must exceed the work function).

## Where the marks are

- A mark for stating the correct starting relationship ($E = hf$ or $E = hc/\lambda$) before substitution.
- A mark for correct unit conversion, such as nm to m or eV to J, shown in the working.
- A mark for the final numerical answer with units and an appropriate power of ten.
- Explanation marks for saying the energy of one photon depends on frequency, not intensity, and that a single photon must exceed the work function.
- Graph marks for drawing the photocurrent curve lower when intensity is reduced, with the stopping potential unchanged for the same frequency.

## Pitfalls

- Forgetting to convert nm to m or eV to J, which gives an answer off by a large power of ten.
- Saying that higher intensity gives each photon more energy, when the energy per photon depends only on frequency.
- Reading the most energetic transition as the one with the highest level number, instead of the largest energy gap.
- Confusing the photon's energy with the electron's kinetic energy in the photoelectric effect, forgetting to subtract the work function.
- Giving an explanation that states light is a particle without linking the observation to the single-photon energy requirement.

## Pattern 9: Photoelectric effect and Planck's constant

Syllabus: S26-U4-SU-31 / S27-U4-SU-31

Frequency: 8 of 10 exam years; 72 marks in all.

Marks reward: Qualitative explanation 19, Final answer 14, Graph work 13, Select the principle or relationship 12, Substitution and working 11, Recall 3

Past questions: 2016P-20(b); 2016P-20(c); 2016P-20(d); 2017P-18(a); 2017P-18(bi); 2017P-18(bii); 2017P-18(ci); 2017P-18(cii); 2017P-18(ciii); 2017P-18(d); 2018P-15(d); 2018R-3(a); 2018R-3(c); 2019P-14(c); 2019P-14(d); 2019R-2; 2021P-13(a); 2021P-13(b); 2021P-13(c); 2021P-13(d); 2022R-1; 2023R-6(a); 2023R-6(b); 2024P-15(b)...

Summary: This point covers the photoelectric effect as an experiment to determine Planck's constant, using the relationship Ek = hf − φ, where φ is the work function of the surface. In the papers it appears mostly as explain questions about the particle model of light, as graph questions reading gradient and intercepts from a kinetic energy versus frequency or energy plot, and as calculations of work function, maximum kinetic energy, stopping voltage and electron speed.

Recognise: Look for photocurrent, ejected or liberated electrons, a metal or metal plate under light, threshold frequency, work function, stopping voltage, or a graph of maximum kinetic energy against frequency or photon energy. Any question asking why brightness or red light does or does not eject electrons is also on this point.

### Steps

1. Write down the given quantities and convert them to consistent units: frequency from wavelength using c = fλ, and energies in eV or J (1 eV = 1.6 × 10^-19 J). Check the question's units for the answer.
2. Apply the photoelectric equation Ek = hf − φ. If hf is less than φ, no electrons are emitted, so the threshold frequency is f0 = φ/h.
3. For a graph of Ek against f, the gradient equals h and the frequency-axis intercept is the threshold frequency. The vertical intercept has magnitude φ, so φ is read from where the line meets the energy axis, not from where it meets the frequency axis.
4. For a graph of Ek against photon energy, the horizontal intercept is the work function, and the gradient is 1 if both axes are in the same units.
5. For stopping voltage, use eVs = Ek(max), so Vs = Ek in eV numerically (or Ek in J divided by e). For electron speed, use Ek = ½mv2 with m = 9.11 × 10^-31 kg, so v = √(2Ek/m) with Ek in joules.
6. For explanations, state the particle model: each photon gives its whole energy to one electron, so a single photon must carry at least φ. Increasing intensity adds more photons, which gives more electrons, but not more energy per electron.

### Where the marks are

- Graph marks are given for labelled axes with units, correctly plotted points, and a line of best fit that is drawn straight through the data.
- Reading the work function from a graph requires the line to be used and the intercept to be taken at the correct axis, with units and sensible significant figures.
- Calculation marks go to showing the eV-to-joule conversion, the correct substitution into the equation, and a final answer with units.
- Explanation marks go to a single-photon, threshold argument and a clear statement that brightness changes the number of electrons but not their maximum kinetic energy.
- Definition marks go to a precise definition of work function as the minimum energy needed to remove an electron from the surface.

### Pitfalls

- Mixing eV and J in the same calculation, or forgetting to convert before substituting into the speed or energy equations.
- Using wavelength where frequency is required in hf, or mixing up the two when reading a question.

- Saying that brightness or intensity increases the energy of each electron, rather than the number of electrons emitted.
- Reading the work function from the wrong intercept on a graph of Ek against frequency, which gives the threshold frequency instead of the work function.
- Giving answers without units or with inconsistent significant figures, or failing to state that a threshold condition must be met.

## Pattern 10: Emission and absorption spectra

Syllabus: S26-U4-SU-32 / S27-U4-SU-32

Frequency: 7 of 10 exam years; 23 marks in all.

Marks reward: Recall 6, Substitution and working 6, Final answer 5, Qualitative explanation 5, Graph work 1

Past questions: 2016R-4(c); 2016R-6; 2017C-21(b); 2018P-14(c); 2019P-14(a); 2019P-14(b); 2021C-20(d); 2021C-20(eii); 2024C-18(c); 2024C-19(c); 2024P-15(a); 2024P-15(ei); 2024P-15(eii); 2025C-20(b); 2025C-20(d)

Summary: This point covers how atoms emit or absorb photons only at the wavelengths matching the energy gaps between their own electron levels, so each element has a line spectrum that acts as a fingerprint for spectral analysis. Papers usually test it through photon energy, frequency and wavelength calculations from an energy-level diagram, plus short explanations of why spectra are discrete and element-specific.

Recognise: Look for an energy-level diagram or a list of levels, a transition named by its levels (for example n = 4 to n = 1), a photon energy, frequency or wavelength to find, or a question asking what a spectrum (bright emission lines, dark absorption lines, a star's spectrum) reveals. Words such as 'unique to the element', 'specific wavelengths' or 'most energetic transition' signal this point.

### Steps

1. Identify the two levels involved: the upper level E2 and the lower level E1. A downward transition emits a photon; an upward transition absorbs one.
2. Find the energy gap with dE = |E2 - E1|, using the level values as given and keeping their signs carefully.
3. Convert to joules if the next step needs them, using 1 eV = 1.60 x 10^-19 J.
4. Find the frequency with f = dE / h, where h = 6.63 x 10^-34 J s.
5. Find the wavelength with lambda = c / f = hc / dE, where c = 3.00 x 10^8 m s^-1.
6. Check the answer: the largest energy gap gives the highest frequency and shortest wavelength, and the value should have sensible powers of ten and units.

### Where the marks are

- A mark for correctly identifying the transition, usually the largest gap or a named pair of levels.
- Marks for the energy difference, often shown in eV before conversion.
- A mark for the joule conversion, and for frequency or wavelength with correct powers of ten and units.
- Explanation marks for linking discrete energy levels to discrete wavelengths, or for stating why each element's lines are unique.

### Pitfalls

- Subtracting the levels in the wrong order or mishandling negative level values, which gives a negative energy gap.
- Leaving the energy in eV when the next formula needs joules.
- Using one level's energy as the photon energy instead of the difference between two levels.
- Choosing the wrong 'most energetic' transition, or misreading a Doppler shift: a source moving toward the observer shifts lines to shorter wavelengths.
- Mixing up emission and absorption, for example describing bright lines when the question is about absorbed photons, or giving a wavelength without its unit.

## Pattern 11: Bohr model of hydrogen

Syllabus: S26-U4-SU-33 / S27-U4-SU-33

Frequency: 8 of 10 exam years; 58 marks in all.

Marks reward: Qualitative explanation 22, Final answer 11, Substitution and working 9, Diagram 6, Select the principle or relationship 4, Recall 3, Graph work 2, Evaluate and conclude 1

Past questions: 2016C-22(a); 2016C-22(b); 2016C-22(ci); 2016C-22(cii); 2016R-6; 2017C-21(b); 2018P-14(a); 2018P-14(b); 2018P-14(c); 2018P-14(e); 2019C-20(d); 2019P-14(a); 2021C-20(d); 2021C-20(ei); 2021R-1; 2022P-15(a); 2022P-15(c); 2022P-15(d); 2024C-19(a); 2024C-19(b); 2024C-19(c); 2025C-20(a); 2025C-20(b); 2025C-20(e)

Summary: This point covers the Bohr model of hydrogen: quantised energy levels explain why atoms emit line spectra, link those lines to energy-level diagrams, and underpin fluorescence, phosphorescence and characteristic X-ray production. Papers usually test it through explanations of discrete spectra, calculations of transition energies, frequencies and wavelengths, and drawing or reading energy-level diagrams, often in mercury, hydrogen, or X-ray and fluorescence contexts.

Recognise: Look for a diagram or list of energy levels with a question asking for the most energetic transition, the frequency or wavelength of the emitted light, or which transitions are possible after a collision with an electron of given energy. Also look for prompts asking why a spectrum consists of sharp lines, why fluorescent radiation identifies an element, how an electron ejected from an inner shell produces an X-ray, or how a graph of wavelength against level numbers gives a constant.

### Steps

1. Identify the upper and lower levels involved and find the energy gap, delta E = E_upper minus E_lower, taking the magnitude for emission.
2. Convert eV to J where needed using 1 eV = 1.60 x 10^-19 J before using photon relationships.
3. For emission, the photon carries delta E = hf = hc/lambda, so f = delta E / h and lambda = hc / delta E. The most energetic transition is the one with the largest energy gap, not necessarily the largest n.
4. For a collision, the atom absorbs only the exact energy of a level gap. The scattered electron keeps E_incident minus delta E; if no excitation occurs the collision is elastic and the electron keeps all its energy.
5. For ionisation or inner-shell X-ray production, the energy needed is the binding energy of the shell. Any excess becomes kinetic energy of the ejected electron: KE = E_incident minus E_binding, and speed follows from KE = 1/2 m v^2.
6. For hydrogen-like ions, energies scale with Z^2 and the Rydberg form is 1/lambda = R Z^2 (1/n1^2 minus 1/n2^2). For characteristic X-rays, Moseley's relation gives sqrt(f) proportional to (Z minus 1), so Z can be found from a measured frequency or wavelength.
7. For fluorescence, describe absorption of a high-energy photon that raises an electron, followed by one or more downward steps each emitting a lower-energy photon. Phosphorescence is the same idea with a delayed emission through a longer-lived state.

### Where the marks are

- Full marks for naming the correct pair of levels and choosing the transition with the largest energy gap rather than the largest quantum number.
- A mark for linking the photon to the energy difference (delta E = hf) and carrying that link through to frequency or wavelength.
- Diagram marks for downward arrows between the correct levels, one mark per transition, with each transition matched to its spectral line.
- Calculation marks for each stage (energy gap, then frequency or wavelength, then unit), so a correct intermediate step still earns marks after a later slip.
- Explanation marks for a complete causal chain: absorbed energy excites the electron, it falls back in steps, and each step emits a photon of definite energy and so a definite wavelength.

### Pitfalls

- Picking the transition with the largest n instead of the largest energy gap, or mixing up the upper and lower levels so the gap has the wrong sign.
- Substituting an energy in eV directly into E = hf or E = hc/lambda without converting to joules.
- Assuming an atom can absorb any electron energy above a level; only the exact level gap is absorbed, so the scattered energies must be worked out case by case.
- Writing a fluorescence explanation that omits the absorption step or describes the emission as a single jump rather than stepwise transitions.
- Mixing up K and L shell energies for X-rays, or ignoring that a larger nuclear charge Z changes the energy levels and so the element-specific wavelengths.

## Pattern 12: Wave–particle duality and de Broglie wavelength

Syllabus: S26-U4-SU-34 / S27-U4-SU-34

Frequency: 9 of 10 exam years; 53 marks in all.

Marks reward: Select the principle or relationship 16, Final answer 12, Substitution and working 9, Qualitative explanation 7, Recall 4, Graph work 4, Units, significant figures, direction 1

Past questions: 2016P-14(a); 2016P-14(b); 2016P-14(c); 2016P-14(d); 2017C-21(aii); 2018P-12(a); 2018P-12(b); 2019R-10(a); 2019R-10(b); 2019R-10(c); 2019R-10(d); 2020R-2; 2022P-13(b); 2022R-9; 2023R-10; 2024C-18(ai); 2024C-19(a); 2024C-19(d); 2025P-13(a); 2025P-13(b); 2025P-13(ci); 2025P-13(cii); 2025P-13(d)

Summary: This point asks students to connect the wave and particle descriptions of photons, electrons and other matter through the de Broglie wavelength (lambda = h/p), the photon relation E = pc, and the double-slit pattern that builds up one particle at a time. It usually appears as a short calculation of wavelength, momentum or speed, followed by an explanation, a graph question or a double-slit fringe-spacing question.

Recognise: Look for a question that gives a particle's mass, speed, kinetic energy, voltage or photon frequency and asks for a wavelength or momentum, or that asks why electrons or photons show interference or diffraction. Graph questions show wavelength against momentum and ask for a linear form, and experimental questions ask about fringe spacing on a detector.

Steps

1. Step 1: List what is given and what is asked. Convert units first: nm to m, eV to J, grams to kg, and check whether the speed is a large fraction of c.
2. Step 2: Find the momentum. For a massive particle at ordinary speeds use p = mv. For a photon use p = h/lambda = E/c = hf/c. If the speed is close to c, use relativistic momentum instead of mv.
3. Step 3: For kinetic energy through a potential difference, find the speed first, for example from qV = (1/2)mv^2, or find p from p = sqrt(2mE), before applying the de Broglie relation.
4. Step 4: Apply lambda = h/p and rearrange for the unknown, such as v = h/(m*lambda). Keep the particle's own mass.
5. Step 5: For double-slit fringe spacing use s = lambda*D/d, with D the screen distance and d the slit separation. A larger speed gives a larger p, a smaller lambda and a smaller fringe spacing.
6. Step 6: For a graph that is not linear, plot lambda against 1/p, since lambda = h/p gives a straight line through the origin whose gradient is Planck's constant.
7. Step 7: For explanations, say the particles behave as waves, name interference or diffraction, and link wavelength to where constructive and destructive regions fall. Equal velocity means equal momentum and so equal wavelength.

Where the marks are

- Stating the relationship lambda = h/p (or h/mv) before substituting earns a method mark even if the final number is wrong.
- Substituting the correct mass for the particle, such as electron or proton mass, and converting units earns a working mark.
- The final answer needs the right unit: m for wavelength, m s^-1 for speed, and kg m s^-1 or N s for momentum.

- Explanations earn marks for naming interference or diffraction and for linking wavelength to fringe positions, not just saying particles act as waves.
- Graph questions earn marks for naming the two variables to plot to get a straight line and for saying the gradient represents Planck's constant.

Pitfalls

- Forgetting to convert nm or eV before calculating, which turns a correct method into a wrong answer.
- Treating a photon as having rest mass and using p = mv, instead of p = h/lambda or hf/c.
- Reporting speed when momentum is asked, or giving momentum without a unit.
- Describing the double-slit change in the wrong direction: a faster electron has a shorter wavelength and so closer fringes.
- Using p = h/lambda for a reflected photon when the question is about recoil; the momentum change on reflection is twice p, giving 2h/lambda.

## Pattern 13: Fluorescence, phosphorescence and X-rays in use

Syllabus: S26-U4-SHE-02 / S27-U4-SHE-02

Frequency: 2 of 10 exam years; 7 marks in all.

Marks reward: Recall 4, Qualitative explanation 3

Past questions: 2018P-14(e)；2021C-20(a)；2021C-20(b)；2021C-20(c)

Summary: This point asks how fluorescence, phosphorescence and X-rays are applied in medicine, forensics, astronomy, industry, lighting and cosmetics, and which property of each makes it suitable. In past papers it appears mostly as short recall of a property or advantage (KNOW), or as a brief explanation of how a material glows or how X-rays reveal hidden detail (EXPLAIN).

Recognise: Look for a question that asks you to list a property of X-rays, give an advantage or application, or explain how a fluorescent material, lamp, label, scanner or telescope works. Key words include UV photons, phosphor coating, fluorescent dye, X-ray spectrometry, airport scanner, food irradiation, space telescope and restoration of artwork.

Steps

1. Identify the process: fluorescence (light re-emitted almost immediately), phosphorescence (light re-emitted over a longer time) or an X-ray application.
2. For a fluorescent material, state that an incoming high-energy photon (for example UV) is absorbed, so an electron is raised to a higher energy level, using E = hf = hc/lambda to show the photon energy is high.
3. The electron drops back down, emitting a photon. Some energy is lost as heat or vibration, so the emitted photon has less energy than the absorbed one and a longer wavelength, which falls in the visible range.
4. For phosphorescence, state that the electron is trapped in a longer-lived (metastable) state, so emission continues after the source is removed.
5. For an X-ray application, name the property that matters: short wavelength and high energy give high penetration (imaging, scanning, irradiation); the characteristic X-rays emitted by the sample have energies set by its elements, which allows X-ray fluorescence to identify composition.
6. Match the property to the use in one sentence, and give each separate advantage its own reason.

Where the marks are

- One mark for stating that absorbed high-energy photons raise electrons to a higher energy level before any light is emitted.
- One mark for explaining that the emitted photon has less energy than the absorbed photon, so it lies in the visible range.
- One mark for each distinct X-ray property named (for example short wavelength, high energy), with a second mark for linking it to the application.

- For a two-advantage question, each advantage must be a different idea, for example one about safety and one about preservation or access.
- Use the terms energy level, photon and absorption/emission correctly; a description of the process in general terms earns the explanation mark only if the cause is stated.

Pitfalls

- Describing the emitted light as having the same energy as the absorbed light, which misses the energy loss that makes the light visible.
- Confusing fluorescence with phosphorescence, or forgetting that phosphorescence has a delay because of a trapped state.
- Writing vague statements like 'X-rays are strong' or 'they are useful' instead of naming a specific property.
- Giving two versions of the same advantage (for example 'safe' and 'harmless'), which earns only one mark.
- Naming a property without connecting it to the application or the question's context.

## Pattern 14: Quantum devices: lasers, photovoltaic cells, LEDs

Syllabus: S26-U4-SHE-03 / S27-U4-SHE-03

Frequency: 2 of 10 exam years; 14 marks in all.

Marks reward: Qualitative explanation 9, Recall 2, Substitution and working 2, Select the principle or relationship 1

Past questions: 2022C-18(a); 2022C-18(b); 2022C-18(c); 2025C-19(b); 2025C-19(g)

Summary: This point asks how quantum physics underlies three devices: the laser, the photovoltaic (solar) cell and the light-emitting diode (LED). It usually appears as a multi-mark 'explain how' or 'describe how' question about energy transitions and charge movement in a pn junction, often paired with a short calculation or comparison using data from a stimulus passage, table or graph.

Recognise: Look for the words PN junction, semiconductor, photon, LED, solar cell, laser, emission or absorption, efficacy or lumens per watt, and an 'explain' or 'describe' command word with 3 to 6 marks. Calculation versions give device data (power per unit, area per unit, voltage, efficacy) in a passage or table and ask for a count, a total area or a current.

Steps

1. Identify the energy change: in an LED, electrons and holes recombine across the junction and release energy as a photon; in a solar cell, an incoming photon is absorbed by an electron in the semiconductor.
2. Link the photon energy to the band gap, using $E = hf = hc/\lambda$, so the colour of emitted light depends on the energy gap between the energy levels.
3. For a solar cell, follow the chain: absorbed photon frees an electron, electron moves from the P-side to the N-side, the layers develop a potential difference, and a closed external circuit lets current flow.
4. For efficiency comparisons, define efficiency as useful output divided by input, and efficacy as luminous flux divided by electrical power (lm $W^{-1}$). Use the graph or passage values to state which device wastes less energy as heat.
5. For power and current questions, use $P = VI$, and luminous flux = efficacy × power, to rearrange for the unknown quantity and substitute the stated values.
6. For area-of-panels questions, find the number of units as total power divided by power per unit, then multiply by area per unit, then convert the unit (for example $m^2$ to $km^2$ by dividing by $10^6$).

Where the marks are

- Explanation marks are given for a correct sequence of events: energy source, photon absorption or emission, movement of charge carriers, then potential difference and current.
- A mark is commonly given for naming the energy change in terms of the energy gap or energy levels,

not just saying electrons 'move' or 'jump'.

- Comparison answers earn one mark for a correct definition or quantity (efficacy or efficiency) and a separate mark for linking it to the supplied graph or data.
- Calculation marks are given separately for pulling each value from the stimulus, for the number of units, for the total area and for the final unit.
- Solar cell explanations need the external circuit stated as the reason current flows.

## Pitfalls

- Describing an LED as 'electrons flowing through it' without the recombination of electrons and holes releasing a photon.
- Confusing efficiency (useful output over input) with efficacy (lumens per watt), so the comparison answer loses marks.
- Reversing the direction of charge movement in a solar cell or forgetting the closed external circuit.
- Using outside values for panel power or area instead of the values stated in the text or table.
- Unit slips in area questions, for example forgetting to convert $m^2$ to $km^2$ or mishandling powers of ten.